

DevOps Lab Program-1

Title:

Build a Docker Container from a Custom Dockerfile

Objective:

To create a custom Docker image using a Flask application and run it inside a container.

Software Requirements:

- Docker
- Python 3
- Flask (specified in requirements.txt)

Files Used:

1. **app.py** – A simple Flask web application.

```
from flask import Flask

app = Flask(__name__)

@app.route('/')
def home():
    return "Docker container running successfully!"

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=9090)
```

Fig-01: The app.py file that includes the statement to be shown

2. requirements.txt

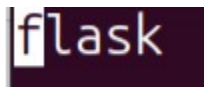


Fig-02: The requirements.txt file listing the required software for installation.

3. Dockerfile

```
#Base image
FROM python:3.9-slim

#Working directory
WORKDIR /app

#Copy files from local system to container
COPY . /app

#Install the dependencies
RUN pip install --no-cache-dir -r requirements.txt || true

#Run the container on port 9090
EXPOSE 9090

#Start the application
CMD ["python","app.py"]
```

Fig-03: The Dockerfile defining the base image, installation and execution commands for dependencies, port configuration, and application startup.

Procedure:

1.Create the Project Directory:

Create a folder named Program-1 and add the following files to it:
app.py, requirements.txt, and Dockerfile.

2.Build the Docker Image:

Open the terminal in the project directory and execute the command:
docker build -t flask-app .

3.Verify Image Creation:

Confirm that the Docker image has been created by running:
docker images

4.Run the Docker Container:

5.Start the container using the command:

docker run -d -p 9090:9090 --name flask-container flask-app

6.Check Active Containers:

7.Verify that the container is running with:

docker ps

Access the Application:

Open a web browser and navigate to:

http://localhost:9090

The message “Docker container running successfully!” should be displayed.

Result:

A custom Docker image was successfully created for the Flask application and deployed in a running container, which is accessible through a web browser.

Conclusion:

This experiment illustrates the process of containerizing a basic Python Flask application using Docker, creating a custom image, and deploying it as a portable and isolated containerized service.